

Retail Business Performance & Profitability Analysis

Elevate Labs Data Analyst Internship — Final Project Report

Introduction

Retail businesses generate large volumes of transactional data, but without analysis, it is difficult to know which product categories are actually driving profit versus which are quietly draining it. This project analyzes a retail superstore's order-level dataset (8,399 transactions, 2009-2012) to identify profit-draining categories, examine whether fulfillment speed affects profitability, and uncover seasonal and regional sales patterns — with the goal of producing actionable business recommendations.

Abstract

Using SQL for aggregation and Python for statistical analysis and visualization, this project cleaned and analyzed retail order data across three product categories (Technology, Office Supplies, Furniture), eight regions, and four seasons. Profit margins were calculated at category, sub-category, and region level. A correlation analysis was run between order fulfillment speed (days to ship, used as an inventory/turnover proxy) and profit margin. Key finding: Furniture is barely profitable (2.3% margin) compared to Technology (14.8%) and Office Supplies (13.8%), driven mainly by heavy losses in Tables and Bookcases. Fulfillment speed showed only a weak correlation with profitability, indicating that pricing and discounting policy — not shipping efficiency — is the primary lever for improving margins.

Tools Used

SQL (SQLite / T-SQL-compatible queries, designed for SSMS): data aggregation, profit margin calculations by category/sub-category/region.

Python (Pandas): data cleaning, feature engineering (days-to-ship, profit margin %, season), correlation analysis.

Python (Matplotlib, Seaborn): visualizations of profit margins, loss drivers, and seasonal trends.

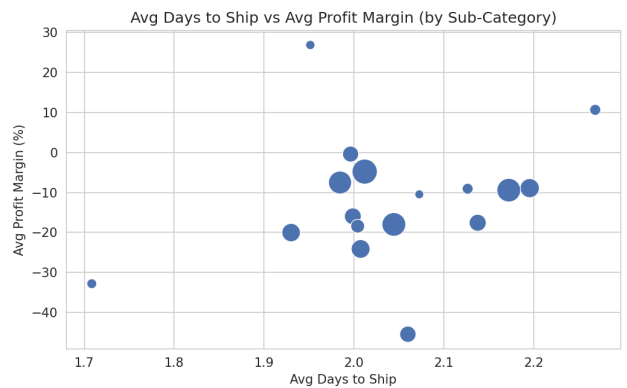
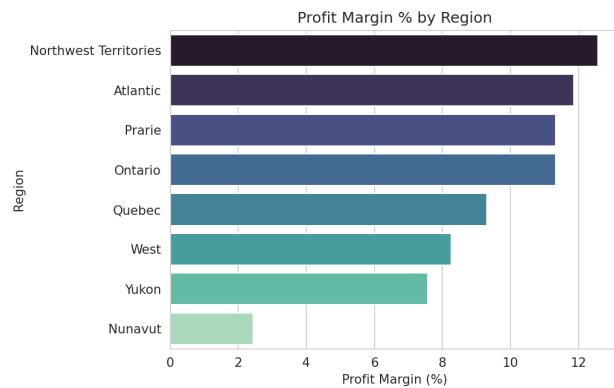
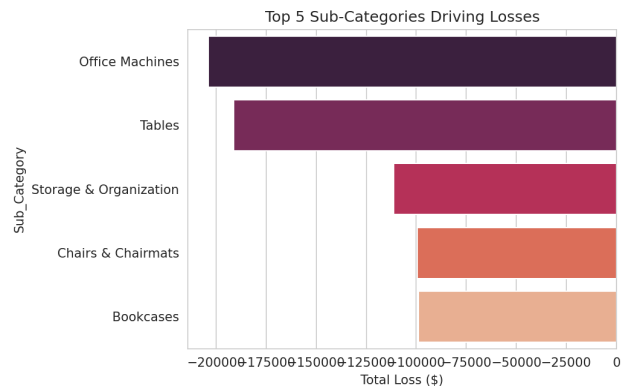
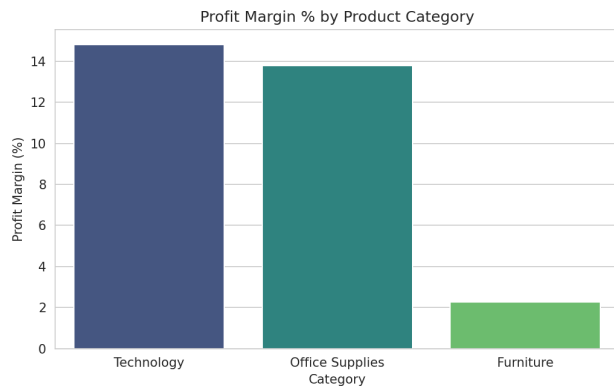
Jupyter Notebook / Google Colab: end-to-end reproducible analysis.

Steps Involved in Building the Project

- Data Import & Cleaning:** Loaded the retail order dataset; filled 63 missing 'Product Base Margin' values with category-level medians; parsed order/ship dates; removed invalid records.
- Feature Engineering:** Derived Profit Margin %, Days to Ship (fulfillment proxy), and Season from Order Date.
- SQL Analysis:** Wrote and ran queries to compute profit margin by category, sub-category, and region, and to identify the top loss-making sub-categories.
- Correlation Analysis:** Tested the relationship between fulfillment speed (days to ship) and profit margin at both order level and sub-category level.
- Visualization:** Built bar charts, scatter plots, and seasonal trend lines to communicate findings clearly.
- Insight Generation:** Translated statistical results into concrete business recommendations.

Key Results

Category	Total Sales	Total Profit	Profit Margin %
Technology	\$5,984,248	\$886,314	14.81%
Office Supplies	\$3,752,762	\$518,021	13.80%
Furniture	\$5,178,591	\$117,433	2.27%



Conclusion

Furniture — specifically Tables and Bookcases — is the clearest profit-draining category, while Technology and Office Supplies remain reliably profitable. Fulfillment speed is not a meaningful driver of profitability (correlation ≈ 0.29 at sub-category level, ≈ 0.01 at order level), so operational shipping improvements alone won't fix margins. The recommended next step is a discount and pricing audit on Furniture sub-categories and underperforming regions (Nunavut, Yukon), alongside a review of whether loss-making SKUs should be repriced, bundled, or discontinued. This analysis demonstrates how SQL aggregation and Python-based statistical testing together can convert raw transactional data into a concrete pricing and inventory strategy.